

SEMESTER 2 EXAMINATIONS 2022/2023

MODULE: CA208 - Logic

PROGRAMME(S):

COMSCI	BSc in Computer Science
CASE	BSc in Computer Applications (Sft.Eng.)
ECSAO	Study Abroad (Engineering & Computing).
ECSA	Study Abroad (Engineering & Computing).

YEAR OF STUDY: 2,O,X

EXAMINERS: Dr. David Sinclair (Internal) (Ext:5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer ALL 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

Section A

QUESTION 1

[Total marks: 30]

Every lion chases some gazelle. Every gazelle who is old is smart. No lion catches any smart gazelle. Any lion who chases some gazelle but does not catch it is hungry.

1(a) [10 Marks]

Convert the statements above into a set of predicate formulae.

1(b) [10 Marks]

Convert the predicate formulae from Question 1(a) into *Prenex Conjunctive Normal Form (PCNF)*.

1(c) [10 Marks]

Using General Resolution, prove that if all gazelles are old then all lions are hungry.

[End Question 1]

QUESTION 2

[Total marks: 10]

2(a) [10 Marks]

Without using truth tables, show that

$$(A \vee B) \rightarrow C \equiv (A \rightarrow C) \wedge (B \rightarrow C)$$

Clearly indicate the logical equivalence used at each step.

[End Question 2]

QUESTION 3

[Total marks: 10]

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)$$

[End Question 3]

QUESTION 4**[Total marks: 10]**

[10 Marks]

Using Gentzen system $\mathcal{G}$ prove

$$(p \rightarrow q) \rightarrow ((p \rightarrow \neg q) \rightarrow \neg p)$$

[End Question 4]**QUESTION 5****[Total marks: 10]**

[10 Marks]

Using *semantic tableaux* prove the following first-order statement is valid.

$$\vdash (\forall x.p(x) \vee \forall x.q(x)) \rightarrow \forall x.(p(x) \vee q(x))$$

Clearly indicate which rule is used at each step.

[End Question 5]**QUESTION 6****[Total marks: 10]**

[10 Marks]

Using Gentzen System $\mathcal{G}$ prove

$$\vdash \forall x.(A \rightarrow B(x)) \rightarrow (A \rightarrow \forall x.B(x))$$

[End Question 6]

Section B

QUESTION 7

[Total marks: 10]

[10 Marks]

Given the following Prolog relations:

- `parent(X,Y)` is true if X is the parent of Y,
- `male(X)` is true if X is male, and
- `female(X)` is true if X is female,

write the following Prolog relations:

- `uncle(X,Y)` is true if X is an uncle of Y,
- `greatuncle(X,Y)` is true if X is the uncle of a parent of Y,
- `ancestor(X,Y)` is true if X is a direct ancestor of Y.

[End Question 7]

QUESTION 8

[Total marks: 10]

[10 Marks]

Let a bus timetable be represented by a list of routes where each route is represented by `Start/End/DepartureTime/ArrivalTime`. Both `DepartureTime` and `ArrivalTime` are represented by a relation `time(Hour,Minutes)`. Write a Prolog relation `journey(Start, End)` that is true if there is a route in the timetable that can get a passenger from `Start` to `End` directly or with one intermediate stop where there is at least 10 minutes between the arrival at an intermediate point the departure to the destination of the journey. Assume the clock is a 24 hour clock and all departure and arrival times occur within the same day

[End Question 8]

QUESTION 9

[Total marks: 10]

[10 Marks]

Describe the procedural interpretation of a Prolog program.

[End Question 9]

APPENDICES

Semantic Tableaux

α	α_1	α_2	β	β_1	β_2
$\neg\neg A_1$	A_1		$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \wedge A_2$	A_1	A_2	$B_1 \vee B_2$	B_1	B_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$	$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$	$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2	$\neg(B_1 \downarrow B_2)$	B_1	B_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$	$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$	$\exists x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$	$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2	β	β_1	β_2
$\neg\neg A$	A		$B_1 \wedge B_2$	B_1	B_2
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \vee A_2$	A_1	A_2	$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$A_1 \rightarrow A_2$	$\neg A_1$	A_2	$\neg(B_1 \uparrow B_2)$	B_1	B_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$	$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2	$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$	$\forall x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$	$\neg\exists x.A(x)$	$\neg A(a)$

[END OF APPENDICES]

[END OF EXAM]